

Abstract

This study evaluates calibration in Groq-hosted Llama 3.3 70B and Gemini 2.0 Flash using 300 saved multiple-choice outputs. Groq reached 0.6733 accuracy and 0.2076 ECE; Gemini reached 0.5767 accuracy and 0.2624 ECE. Both models showed overconfidence, with incorrect-answer confidence of 0.8613 for Groq and 0.8680 for Gemini.

Introduction

Reliable uncertainty estimates are important when LLM answers are used in educational, professional, or decision-support contexts. This project asks whether two deployed LLMs know when they are wrong, using self-reported confidence and empirical correctness.

Related Work

Prior calibration work introduced reliability diagrams, ECE, and temperature scaling for classifiers. Recent LLM work studies whether language models can express uncertainty in words. This project differs by applying these tools to saved Gemini and Groq API outputs with category-level analysis and paired significance testing.

Research Questions

RQ1: How accurate and calibrated are Groq and Gemini on this benchmark? RQ2: Do model differences appear statistically significant? RQ3: Which question categories and error types explain the observed calibration failures?

Methodology

The analysis uses the saved CSV outputs only. Questions are classified into Factual Knowledge, Mathematical Reasoning, and Logical Reasoning using deterministic rules over dataset categories and question wording. Metrics include accuracy, average confidence, ECE, and overconfidence.

Experimental Setup

The dataset contains 300 questions answered by both models. Statistical comparison uses exact McNemar's test for paired correctness, a chi-square test over accuracy counts, and a paired bootstrap with 10000 resamples.

Results

Overall metrics:

model	category	n_questions	accuracy	average_confidence	ece	overconfidence
groq	All Questions	300	0.6733	0.8769	0.2076	0.8613
gemini	All Questions	300	0.5767	0.8391	0.2624	0.8680

Category metrics:

model	category	n_questions	accuracy	average_confidence	ece	overconfidence
groq	Factual Knowledge	228	0.6754	0.8830	0.2103	0.8654
groq	Mathematical Reasoning	39	0.6410	0.8604	0.2194	0.8474
groq	Logical Reasoning	33	0.6970	0.8546	0.2175	0.8498
gemini	Factual Knowledge	228	0.5921	0.8452	0.2530	0.8792
gemini	Mathematical Reasoning	39	0.5128	0.8235	0.3107	0.8325
gemini	Logical Reasoning	33	0.5455	0.8159	0.2821	0.8439

McNemar $p=0.027024$; chi-square $p=0.018218$; bootstrap accuracy-difference CI=(0.0167, 0.1767).

Error Analysis

Incorrect responses were categorized as hallucination, calculation error, logical error, incomplete answer, or confidence miscalibration.

model	error_type	count
gemini	Calculation Error	12
gemini	Confidence Miscalibration	77
gemini	Hallucination	27
gemini	Logical Error	11
groq	Calculation Error	6
groq	Confidence Miscalibration	63
groq	Hallucination	26
groq	Logical Error	3

Threats to Validity

The dataset is curated and compact, category sizes are uneven, confidence is self-reported text rather than logits, and the model comparison covers only two provider snapshots. Multiple-choice scoring also hides partial knowledge and explanation quality.

Discussion

Groq performs better on this saved benchmark, but both models are miscalibrated. High confidence is not sufficient evidence of correctness. Category-specific diagrams are more informative than an overall reliability plot alone.

Conclusion

The project now supports a stronger empirical claim: Groq outperforms Gemini on these saved outputs, both models are overconfident, and category/error analyses identify where calibration failures concentrate.

Future Work

Future work should use larger public benchmarks, repeated runs, log-probability confidence where available, human confidence baselines, and more models. Confidence elicitation prompts should also be varied systematically.